$$X_t \in \mathbb{R}^{4096\times 256}, \quad M \in \mathbb{R}^{K\times 64}$$

$$Q = X_t W_Q,\ K_M = M W_K,\ V_M = M W_V$$

$$\mathrm{Attn}(X_t, M) = \mathrm{softmax}\left(\frac{Q K_M^\top}{\sqrt{d}}\right) V_M$$